

EDUCATION	Ph.D. Electrical and Computer Engineering <i>University of Delaware</i> • Advisor: Dr. Xing Gao, Dr. Chase Cotton • Dissertation: Red-teaming LLMs: Bypassing security guardrails of AI-driven systems by hiding malicious intent in prompts. Converted unsafe prompt to safe by token manipulation.	2019 - 2026 (<i>expected</i>) Newark, DE
	M.S. Cybersecurity <i>University of Delaware</i> • Advisor: Dr. Chase Cotton • Thesis: Taxonomy of network traffic to perform ML-based anomaly detection for new attack signals	2017 - 2019 Newark, DE
TECHNICAL SKILLS	Network Security: Packet analysis, Network traffic analysis, Pen testing, Reverse engineering, System hardening, Cryptography Machine Learning: Malware detection, network traffic classification, vulnerability detection and repair, LLM safety, Natural Language Processing (NLP), Statistical Analyses ML Frameworks & Tools: Torch, TensorFlow, Scikit-learn, Keras, Hugging Face, NumPy Programming Languages: Python, R, MATLAB, Bash, Java, C++, MySQL	
PUBLICATIONS	Code Vulnerability Repair with LLM using Context-Aware Prompt Tuning <i>IEEE Security and Privacy Workshops 2025</i> Improved buffer overflow vulnerability repair by 74% on GitHub Copilot. Used prompt tuning to introduce security and code knowledge to code-generating LLMs. Ensemble Defense System: A Hybrid IDS Approach for Effective Cyber Threat Detection <i>IEEE International Telecommunication Networks and Applications Conference 2023</i> Developed a hybrid intrusion detection model by integrating signature-based and anomaly-based IDS tools. Designed an interactive tracker for tracking and visualizing threat vectors. Efficient attack detection in IoT devices using feature engineering-less machine learning <i>International Journal of Computer Science and Information Technology 2022</i> Designed a lightweight malware detection model that removed the feature-engineering process from the ML pipeline. Deployable on battery-powered, low-memory IoT devices. Detecting Attacks on IoT Devices using Featureless 1D-CNN <i>IEEE International Conference on Cyber Security and Resilience 2021</i> Used pcap files as raw time-series signals to train a 1D-CNN for malware detection. Emphasize the role of headers in malware detection. A Feature Taxonomy for Network Traffic <i>University of Delaware, Department of Electrical and Computer Engineering 2019</i> Developed a taxonomy model to detect anomalous activities detected in packets, flows, and sessions. Performed extensive analysis of ML-based anomaly detection techniques based on raw, behavioral, and temporal features. Highlighted anomalies detected by network traffic scanners such as Zeek and Wireshark.	
PROJECTS	JPMC Data for Good Hackathon 2021 Led a team of 4 to design a forecasting model with Multinomial Naive Bayes with a focus on sparse data handling. Twitter Bot Detection and Classification 2019 Built an NLP classification model, achieving 91% accuracy on detecting harmful bot accounts on Twitter and classifying them into political, social, and personal harm categories.	

WORK
EXPERIENCE

Graduate Research Assistant | University of Delaware 2019 – Present

- Developed and taught a graduate-level class on implementing supervised ML techniques for anomaly detection in large-scale data sources like packet captures or compiled network security logs from Zeek.
- Collaborated with professors to guide undergraduate, graduate, and mid-career professionals as a TA. Resolve technical and conceptual difficulties faced by students in conducting ML experiments, building end-to-end ML pipelines for anomaly identification.

Backend & Analytics Developer | Women Plus Blockchain 2018.06 - 2018.08

- Built 50 web crawlers to collect data from blockchain news outlets; implemented a scalable pipeline on AWS console to filter relevant articles using SEO-based screening
- Boosted subscriber growth by 17% by automating user engagement metrics on Google Analytics dashboard in 4 months

Systems Engineer | Tata Consultancy Services Ltd 2014.09 - 2017.04

- Led an application team of 5 members to support and maintain a telecom banking application, serving daily transactions
- Deployed changes to the production environment after application testing, resolved critical incidents, maintaining 99% uptime SLA
- Cut reporting costs by 90% by automating daily business intelligence report generation using VBA

TEAMWORK
EXPERIENCE

- Led peer mentoring organization EmPOWER at University of Delaware 2021-2023
- Associate, Hispanic Latino Graduate Student Organization 2022
- Treasurer, Indian Graduate Student Organization 2021